

ENCODER

Input Image

$3 \times H \times W$

Stem

Conv $3 \times 3 \times 2$, C=48

ConvNeXt $\times 2$

C = 48, H $\times$ W

$\downarrow 2\times$

ConvNeXt $\times 2$

2C = 96, H/2

$\downarrow 2\times$

ConvNeXt $\times 2$

4C = 192, H/4

$\downarrow 2\times$

Hamiltonian Bottleneck $\times 2$

ConvNeXt || Oscillator Bank $\rightarrow$ Learned Gate + Dropout | 8C = 384, H/8
outputs: features f · momentum p · energy H

DECODER

Output Mask

$N \times H \times W$

Seg Head

Conv 3×3 + Conv 1×1

E-skip + Mom + CNX $\times 2$

C = 48, H $\times$ W

$\uparrow 2\times$

E-skip + Mom + CNX $\times 2$

2C = 96, H/2

$\uparrow 2\times$

E-skip + Mom + PS attn

+ ConvNeXt $\times 2$, 4C = 192, H/4

Energy H (gating)

e_1

e_2

e_3

skip (energy-gated)

$\tilde{e} = \sigma(\gamma \cdot (H - H)) \odot e$

skip (energy-gated)

skip (energy-gated)

d_1

d_2

d_3

Momentum p

BOTTLENECK DETAIL

ConvNeXt

gradient flow

Oscillator Bank (4-dir)

$z = e^{(-\nu + i\omega)\Delta} \cdot z + u \rightarrow q, p, H$

Gate + Drop

$g = \sigma(W[f_c; f_o])$

$\rightarrow$ f (features)

$\rightarrow$ p (momentum)

$\rightarrow$ H (energy)



Standard (Encoder)



Hamiltonian (Decoder)



Skip connection



Energy H



Momentum p



Data flow